

M:N Relationship Relations

Book_Author

BookID (FK)

Book_Publisher

Many to many shown in ERD requires junction table

BookID (FK)

Normalization Steps From First To Third Normal Form

Step 1: 1NF

A Table Meets First Normal Form

We decompose:

Membership Name First Name Last Name

Staff Name Split Into First And Last

Every connection sits in first normal form. To fully satisfy 1NF, the multivalued attribute Genre in the book table is decomposed into a separate relation: Book_Genre (Book_ID,Genre).

Step 2: 2NF

A Table Meets Second Normal Form

Composite primary keys appear exclusively in many-to-many relationship tables

Book_Author(BookID, AuthorID)

Book_Publisher(BookID, PublisherID)

With nothing beyond key fields here, partial ties simply do not exist. These setups stay free of extra bits, leaving zero room for split reliance.

All Tables Now in Second Normal Form.

Step 3: 3NF

Check each table:

Membership

Loan

Others

Final Relations In Third Normal Form

MEMBERSHIP Membership with unique ID includes; MembershipID, first name, last name, phone Number, DateOfBirth and StaffID.

STAFF

Staff with ID first name, last name

FINE

Fine FineID-> MembershipID, Amount, Reason, DateAssessed, Paid

BOOK

Book with unique ID title and genre

Book_Genre (BookID, Genre)

BOOKCOPY

Book Copy Details

AUTHOR

Author with unique ID first name, last name

PUBLISHER

Publisher(PublisherID PK, Name)

LOAN

Loan loanID, MembershipID, CopyID, StaffID, LoanDate, DueDate, ReturnDate, OverdueAmount.

RESERVATION

Reservation ID links membership and book with reservation date

BOOK_AUTHOR (M:N)

Book Author Book ID Foreign Key Author ID Foreign Key Primary Key Book ID Author ID

BOOK_PUBLISHER (M:N)

Book Publisher Relationship With Book ID And Publisher ID As Primary Key